ESP 600 – ELECTROSTATIC PRECIPITATOR

INTENDED USE

- VANTERM ESP-600 electrostatic precipitators are used to filter the oily dust, smoke and odor emissions generated in a stenter frames in the textile industry.
- Thanks to the integrated heat exchanger module, the ESP-600 units provide an excellent heat recovery and saves energy, as well. This is a very important advantage for the filtration of hot exhaust gas emissions created in the stenter frames.
- ESP-600 units are also used for filtration of smoke, dust and sticky-greasy aerosoles that are generated from aluminum die-casting processes.

ENERGY SAVING

- The heat exchanger module uses water and transfers the heat energy of the hot gas emissions to water.
- VANTERM heat exchanger used in ESP-600 precipitators are made of 304-quality stainless steel.
- Thanks to its perfect design and high heat transfer efficiency, the water temperature can rise up to 40-65°C, on a typical stenter application, depending on the water volume and stenter capacity.
- The gained heat energy can be used in different processes within the plant and provides energy saving.
- Since the energy costs have a great impact on the production cost of the stenter machines, the saved energy by the ESP-600 brings a very valuable advantage for the textile industry.



Heat exchanger with high efficiency.



Electrostatic filter cassettes

ESP 600 FILTRATION TECHNOLOGY

- VANTERM ESP-600 electrostatic precipitator has an ioniser stage which Works in 12 000 Volts. This high voltage ionizes the dust-fume and smoke particles in the dirty air path.
- After passing through the ioniser stage, the dirty air reaches the collector plates that separates the particles effectively.
- There are also mesh filter stages inside the ESP-600 precipitators that enhance the filtration quality. As a result, the cleaned air can be exhausted without any problem to the atmosphere.
- VANTERM ESP-600 electrostatic precipitators have been tested successfully with Brückner stenter machines according to the EN TS 13725 standard. The results were highly satisfactory as 270 - 300 KB/m³.
- Thanks to the intelligent control via 10.1” colored-touch screen with PLC system, the operation of the ESP-600 is very easy and user-friendly.
- All the functions of the system can be controlled over the touch screen and also the control can be done remotely over a network from another control.
- The filter unit has also an automatic cleaning which is triggered easily over the touch screen.

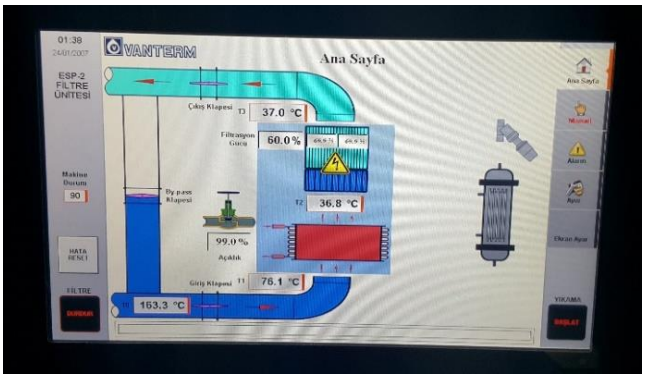
CE

EASE OF USE & ASSEMBLY

- Easy to assemble at site, due to its modular design.
- Easy maintenance and service, thanks to its automatic filter cleaning system.
- The heat exchangers and the electrostatic precipitators are cleaned effectively at the same time.
- It is enough for the operator just to check the level of the water and the chemical cleansing agent.
- During the automatic cleaning period, the water is heated up to a certain temperature automatically, through the integrated heat exchanger. (with steam or electric power.)
- Easy to drain the dirty water after finishing the cleaning period.
- The cleaning period is started-paused or stopped easily by the operator via 10.1 colored-touch screen.
- ESP-600 electrostatic precipitators have the dirty-air inlet and clean-air outlet on the left-hand side of the filter unit, as standard. On request, the position of air-inlet and air-outlet can be produced on the right-hand side of the filter unit, depending on the system lay-out.

SAFETY & DURABILITY

- VANTERM ESP-600 electrostatic precipitator is made of 304.quality stainless steel.
- Very durable against corrosion and wear.
- Thanks to the low pressure loss of the ESP-600 precipitator, in most cases the own-ventilators of the stenter frames are enough for an effective extraction system.
- The optional fire-sensing and fire-alarm system can also integrated in the VANTERM ESP-600 precipitator, on request.
- Easy and user-friendly maintenance and service, thanks to the hinged/locked access doors from front and back-side of the filter unit.



10 inch colored-touch screen control panel



Heat exchanger for cleaning water.



AKDEM Textile Company – Kestel -Bursa
Brückner stenter frame with 8 chambers
Energy saving is about 25-30 MWh (25000-30000 kWh) per month.

***** Ongoing product development may lead to change without notice.**

VANTERM Isı Ve Makina San. Tic. A.Ş.
GOSB 800. Sok. No:804 41480 Gebze/Kocaeli TÜRKİYE
Tel:+90 262 751 05 94(Pbx) www.vanterm.com

VANTERM